Integration by Parts

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The integration by parts formula**

Evaluate $\int xe^x\,dx$ using integration by parts.

Let $u = x$, $dv = e^x\,dx$: $= xe^x - \int e^x\,dx = xe^x - e^x + C$.

2 **[Foundational]** • **Target: Repeated and definite applications**

Evaluate $\int x^2e^x\,dx$, applying integration by parts twice.

First pass ($u = x^2$): $x^2e^x - 2\int xe^x\,dx$. Second pass gives $\int xe^x\,dx = xe^x - e^x$. Total: $x^2e^x - 2xe^x + 2e^x + C$.

3 **[Foundational]** • **Target: Repeated and definite applications**

Evaluate $\int_0^1 xe^{2x}\,dx$ exactly.

Let $u = x$, $dv = e^{2x}dx$: $\left[\frac{x}{2}e^{2x} - \frac{1}{4}e^{2x}\right]_0^1 = \left(\frac{e^2}{2} - \frac{e^2}{4}\right) - \left(0 - \frac{1}{4}\right) = \frac{e^2}{4} + \frac{1}{4}$.

4 **[Foundational]** • **Target: Choosing u and dv wisely (LIATE)**

Evaluate $\int x\ln x\,dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as u under the LIATE rule).

$u = \ln x$, $dv = x\,dx$, so $du = \frac{1}{x}dx$, $v = \frac{x^2}{2}$: $= \frac{x^2}{2}\ln x - \int \frac{x^2}{2}\cdot\frac{1}{x}\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$.

5 **[Foundational]** • **Target: The recurring-integral trick**

Evaluate $\int e^x\sin x\,dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.

Let $I = \int e^x\sin x\,dx$. First pass ($u = \sin x$, $dv = e^x dx$): $I = e^x\sin x - \int e^x\cos x\,dx$. Second pass on the new integral ($u = \cos x$): $\int e^x\cos x\,dx = e^x\cos x + \int e^x\sin x\,dx = e^x\cos x + I$. So $I = e^x\sin x - e^x\cos x - I$, giving $2I = e^x(\sin x - \cos x)$, so $I = \frac{e^x(\sin x - \cos x)}{2} + C$.

6 **[Foundational]** • **Target: Reduction formulas**

Use integration by parts with $u = x^n$, $dv = e^x dx$ to derive the reduction formula $\int x^n e^x\,dx = x^n e^x - n\int x^{n-1}e^x\,dx$.

$du = nx^{n-1}dx$, $v = e^x$: $\int x^n e^x\,dx = x^n e^x - \int e^x \cdot nx^{n-1}\,dx = x^n e^x - n\int x^{n-1}e^x\,dx$.

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7 [Foundational] • Target: Reduction formulas

Use the reduction formula above (applied twice) to evaluate $\int x^2 e^x\,dx$, confirming it matches an earlier result.

$\int x^2 e^x\,dx = x^2e^x - 2\int xe^x\,dx$. Applying again: $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x$. Total: $x^2e^x - 2(xe^x - e^x) = x^2e^x - 2xe^x + 2e^x + C$, matching the earlier double-integration-by-parts result.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration by parts with inverse trig functions

Evaluate $\int x\sin^{-1}x\,dx$ using $u = \sin^{-1}x$, $dv = x\,dx$ (the resulting integral can be left in terms of a known antiderivative form).

$du = \frac{1}{\sqrt{1-x^2}}dx$, $v = \frac{x^2}{2}$: $= \frac{x^2}{2}\sin^{-1}x - \frac{1}{2}\int \frac{x^2}{\sqrt{1-x^2}}\,dx$, which requires a further trigonometric substitution to complete.

9 [Intermediate] • Target: A capstone mixed-technique problem

Evaluate $\int_0^{\pi/2} x\cos(2x)\,dx$ exactly, using integration by parts.

$u = x$, $dv = \cos(2x)dx$, $v = \frac{1}{2}\sin(2x)$: $\left[\frac{x}{2}\sin(2x)\right]_0^{\pi/2} - \frac{1}{2}\int_0^{\pi/2}\sin(2x)\,dx = \left(\frac{\pi}{4}\sin\pi - 0\right) - \frac{1}{2}\left[-\frac{1}{2}\cos(2x)\right]_0^{\pi/2} = 0 - \frac{1}{2}\left(\frac{1}{2} + \frac{1}{2}\right) = -\frac{1}{2}$.

10 [Intermediate] • Target: Visualizing an area found by integration by parts

The graph shows $f(x) = xe^{-x}$ on $[0, 3]$. Find the exact area under the curve on this interval using integration by parts.

$u = x$, $dv = e^{-x}dx$, $v = -e^{-x}$: $\left[-xe^{-x}\right]_0^3 + \int_0^3 e^{-x}\,dx = -3e^{-3} + \left[-e^{-x}\right]_0^3 = -3e^{-3} + (1 - e^{-3}) = 1 - 4e^{-3} \approx 0.801$.

11 [Intermediate] • Target: Choosing between substitution and parts

For each integral, state whether substitution or integration by parts is the appropriate technique, and identify the key choice (u-substitution variable, or u/dv for parts) without fully evaluating: (a) $\int xe^{x^2}\,dx$; (b) $\int xe^{2x}\,dx$.

(a) Substitution — $u = x^2$ makes $du = 2x\,dx$ appear directly, no product rule needed. (b) Integration by parts — x and e^{2x} are genuinely different function types with no simplifying substitution; use $u = x$, $dv = e^{2x}dx$.

12 [Intermediate] • Target: Choosing between substitution and parts

Evaluate $\int \sec^3 x\,dx$ using integration by parts with $u = \sec x$, $dv = \sec^2 x\,dx$, and the identity $\tan^2 x = \sec^2 x - 1$ to solve the resulting equation for the original integral.

$du = \sec x\tan x\,dx$, $v = \tan x$: $I = \sec x\tan x - \int \sec x\tan^2 x\,dx = \sec x\tan x - \int \sec x(\sec^2 x - 1)\,dx = \sec x\tan x - \int \sec^3 x\,dx + \int \sec x\,dx$. So $2I = \sec x\tan x + \ln|\sec x + \tan x|$, giving $I = \frac{1}{2}(\sec x\tan x + \ln|\sec x + \tan x|) + C$.

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13 [Intermediate] • Target: Choosing between substitution and parts

A city's population growth rate is modeled by $r(t) = t\ln(t+1)$ thousand people per year, for $0 \le t \le 5$. Set up and evaluate the definite integral for the total population growth over these 5 years using integration by parts, giving your answer to two decimal places.

Total growth $= \int_0^5 t\ln(t+1)\,dt$. This does not integrate cleanly with $u = \ln(t+1)$, $dv = t\,dt$ alone; using $u = \ln(t+1)$, $dv = t\,dt$, $v = \frac{t^2}{2}$: $\left[\frac{t^2}{2}\ln(t+1)\right]_0^5 - \frac{1}{2}\int_0^5 \frac{t^2}{t+1}\,dt$. Dividing $\frac{t^2}{t+1} = t - 1 + \frac{1}{t+1}$: $\frac{1}{2}\int_0^5 (t - 1 + \frac{1}{t+1})dt = \frac{1}{2}\left[\frac{t^2}{2} - t + \ln(t+1)\right]_0^5 = \frac{1}{2}(12.5 - 5 + \ln 6) \approx 4.65$. So total growth $\approx \frac{25}{2}\ln 6 - 4.65 \approx 22.41 - 4.65 \approx 17.76$ thousand peop.

14 [Intermediate] • Target: The recurring-integral trick

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x\,dx$ for integers $n \ge 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \ge 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x\,dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n\to\infty} I_n$.

(a) Base integral: $I_0 = \int_0^1 e^x\,dx = [e^x]_0^1 = e^1 - e^0 = e - 1$. (b) Reduction formula via IBP: Let $u = x^n \Rightarrow du = nx^{n-1}\,dx$. Let $dv = e^x\,dx \Rightarrow v = e^x$. $I_n = [x^n e^x]_0^1 - n\int_0^1 x^{n-1}e^x\,dx = (1^n e^1 - 0) - nI_{n-1} = e - nI_{n-1}$. ■ (c) Compute I_3: - $I_1 = e - 1(I_0) = e - (e-1) = 1$. - $I_2 = e - 2(I_1) = e - 2(1) = e - 2$. - $I_3 = e - 3(I_2) = e - 3(e-2) = e - 3e + 6 = 6 - 2e$. (d) Squeeze Theorem & Limit: For.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Definite integrals by parts • Cross-Lesson: 2.02 Continuity

Level 3 · Advanced / AP FRQA particle moves along a planar path with velocity $\vec{v}(t) = \langle \sin(t^2), \cos(t^3) \rangle$ for $0 \le t \le 3$. Write the definite integral representing the total distance traveled by the particle, and calculate the numerical value using three decimal places.

Written Setup: Total Distance $= \int_0^3 \sqrt{\sin^2(t^2) + \cos^2(t^3)}\,dt$. Numerical Evaluation: Entering this into a graphing calculator yields: Total Distance $\approx$ **2.874** (or 2.873).

16 [Challenging] • Target: Reduction formulas • Cross-Lesson: 3.01 The Derivative from First Principles

[Unit 8 · Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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17 **[Challenging]** • **Target: Integration by parts with inverse trig functions** • **Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

Level 3 · Advanced / AP FRQEvaluate the improper integral $\int_0^\infty xe^{-2x}\,dx$ or establish that it diverges.

Rewrite with limit notation: $\lim_{b\to\infty}\int_0^b xe^{-2x}\,dx$. Use integration by parts: $u=x\Rightarrow du=dx$, $dv=e^{-2x}\,dx\Rightarrow v=-\frac{1}{2}e^{-2x}$. $\int xe^{-2x}\,dx=-\frac{1}{2}xe^{-2x}-\int\left(-\frac{1}{2}e^{-2x}\right)dx=-\frac{1}{2}xe^{-2x}-\frac{1}{4}e^{-2x}$. Evaluate from 0 to b: $\lim_{b\to\infty}\left[-\frac{1}{2}be^{-2b}-\frac{1}{4}e^{-2b}-\left(0-\frac{1}{4}e^{0}\right)\right]$. By L'Hôpital's Rule.

18 **[Challenging]** • **Target: A capstone mixed-technique problem** • **Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals**

Level 2 · MediumFind the equations of all horizontal and vertical tangent lines to the ellipse $4x^2+9y^2=36$.

Differentiate implicitly: $8x+18y\frac{dy}{dx}=0\Rightarrow\frac{dy}{dx}=-\frac{8x}{18y}=-\frac{4x}{9y}$. 1. Horizontal tangents (numerator = 0): $-4x=0\Rightarrow x=0$. Substitute into curve: $4(0)^2+9y^2=36\Rightarrow y^2=4\Rightarrow$ **$y=\pm2$**. 2. Vertical tangents (denominator = 0): $9y=0\Rightarrow y=0$. Substitute into curve: $4x^2+9(0)^2=36\Rightarrow x^2=9\Rightarrow$ **$x=\pm3$**.

19 **[Challenging]** • **Target: Visualizing an area found by integration by parts** • **Cross-Lesson: 4.02 Riemann Sums and Definite Integrals**

Level 2 · MediumThe base of a solid is bounded by $y=\sqrt{x}$, the line $x=9$, and the x-axis. Cross sections perpendicular to the x-axis are equilateral triangles. Find the volume of the solid.

Base side length: $s(x)=\sqrt{x}-0=\sqrt{x}$. Equilateral triangle area formula: $A(x)=\frac{\sqrt{3}}{4}[s(x)]^2=\frac{\sqrt{3}}{4}(\sqrt{x})^2=\frac{\sqrt{3}}{4}x$ Integrate from $x=0$ to $x=9$: $V=\int_0^9\frac{\sqrt{3}}{4}x\,dx=\frac{\sqrt{3}}{4}\left[\frac{x^2}{2}\right]_0^9=\frac{\sqrt{3}}{4}\cdot\frac{81}{2}=\mathbf{\frac{81\sqrt{3}}{8}}\approx17.537$.

20 **[Challenging]** • **Target: Choosing between substitution and parts** • **Cross-Lesson: 5.01 The Fundamental Theorem of Calculus**

Level 1 · FoundationsEvaluate the indefinite integral $\int\arctan(x)\,dx$.

Use LIATE: $u=\arctan(x)\Rightarrow du=\frac{1}{1+x^2}\,dx$, and $dv=dx\Rightarrow v=x$. Applying parts: $\int\arctan(x)\,dx=x\arctan(x)-\int\frac{x}{1+x^2}\,dx$. For the second integral, let $w=1+x^2\Rightarrow dw=2x\,dx$: $\int\frac{x}{1+x^2}\,dx=\frac{1}{2}\ln(1+x^2)$. Thus: $\mathbf{x\arctan(x)-\frac{1}{2}\ln(1+x^2)+C}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The integration by parts formula

Using the reduction formula, evaluate $\int_0^{\frac{\pi}{2}} \cos^6(x)\,dx$.

a. $\frac{5\pi}{16}$

b. $\frac{5\pi}{32}$ ✓

c. $\frac{3\pi}{16}$

d. $\frac{1}{6}$

On $[0, \pi/2]$, $I_n = \frac{n-1}{n} I_{n-2}$ with $I_0 = \pi/2$ and $I_1 = 1$. Applying it repeatedly gives $\frac{5\pi}{32}$.

22 [Foundational] • Target: Repeated and definite applications

Reduction formula: $I_n = \int \sin^n x\,dx$ satisfies:

a. $I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$ ✓

b. $I_n = \frac{\sin^{n+1} x}{n+1} + C$

c. $I_n = -n\cos x \sin^{n-1} x + C$

d. $I_n = I_{n-1} \cdot I_1$

Standard reduction: $I_n = -\frac{\cos x \sin^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$.

23 [Foundational] • Target: Choosing u and dv wisely (LIATE)

Volume between paraboloids $z = x^2 + y^2$ and $z = 4 - x^2 - y^2$:

a. 4π

b. 8π ✓

c. 16π

d. $32\pi/3$

Intersect at $r = \sqrt{2}$; $V = 2\pi \int_0^{\sqrt{2}} r(4 - 2r^2) dr = 8\pi$.

24 [Foundational] • Target: The recurring-integral trick

Consumer surplus formula (demand $p = D(q)$, equilibrium (q^*, p^*)):

a. $\int_0^{q^*} [D(q) - p^*]\,dq$ ✓

b. $\int_0^{p^*} D(q)\,dq$

c. $p^* q^* - \int_0^{q^*} D(q)\,dq$

d. $D(q^*) - p^*$

Consumer surplus = area above price, below demand curve.

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25 [Foundational] • Target: Definite integrals by parts

Reduction formula for $\int \sin^n x\,dx$ comes from IBP with $u =$

a. $\sin^n x$

b. $\cos x$

c. $\sin^{n-1} x$ ✓

d. x

Write $\sin^n x = \sin^{n-1} x \cdot \sin x$, then IBP.

26 [Foundational] • Target: Reduction formulas

Tabular integration (repeated parts): $\int x^3 e^x\,dx$. Number of IBP steps needed:

a. 2

b. 1

c. 4

d. 3 ✓

x^3 differentiates to 0 in 3 steps.

27 [Foundational] • Target: Integration by parts with inverse trig functions

Archimedes' principle as an integral: buoyancy force = weight of fluid displaced = ρg times:

a. surface area

b. **volume of submerged body** ✓

c. pressure × depth

d. mass of body

Buoyancy = weight of displaced fluid.

28 [Foundational] • Target: A capstone mixed-technique problem

Surface area of revolution about x-axis: $S =$

a. $\pi\int_a^b f(x)\sqrt{1+f'(x)}\,dx$

b. $2\pi\int_a^b f(x)\sqrt{1+[f'(x)]^2}\,dx$ ✓

c. $2\pi\int_a^b [f'(x)]^2\,dx$

d. $\int_a^b f(x)dx$

Surface area of revolution formula.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Visualizing an area found by integration by parts

Area of surface of revolution $r = f(\theta)$ about polar axis:

a. $2\pi\int f\sin\theta\sqrt{f^2 + f'^2}\,d\theta$ ✓

b. $\pi\int f^2\,d\theta$

c. $2\pi\int f\,d\theta$

d. $4\pi\int f\cos\theta\,d\theta$

Surface area in polar coordinates.

30 [Intermediate] • Target: Choosing between substitution and parts

Gauss-Green formula in 2D: $\iint_\Omega abla\cdot\mathbf{F}\,dA =$

a. $\oint_{\partial\Omega}\mathbf{F}\cdot\mathbf{n}\,ds$ ✓

b. $\oint_{\partial\Omega}\mathbf{F}\cdot d\mathbf{s}$

c. $\iint_\Omega|\mathbf{F}|\,dA$

d. 0

Green's theorem (divergence form).

31 [Intermediate] • Target: The integration by parts formula

$\int\sin^4 x\,dx$ using reduction formula:

a. $\frac{3x}{4} + \frac{\sin 2x}{4} + C$

b. $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$ ✓

c. $\frac{x}{2} - \frac{\sin 4x}{8} + C$

d. $\frac{\sin^3 x\cos x}{4} + \frac{\sin x\cos x}{4} + \frac{x}{8} + C$

Double application of half-angle.

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32 [Intermediate] • Target: Repeated and definite applications

The graph of a function $f'(x)$ is shown below. It is given that $f(x)$ is a function such that $f(1) = 5$. The area between the curve $y = f'(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 5$ is represented by the shaded region. If the area of the shaded region is 16 square units, determine the value of $f(5)$.

a. 17

b. **21** ✓

c. 9

d. 11

The area of the shaded region is given by the definite integral of $f'(x)$ from $x = 1$ to $x = 5$. This area is also equal to the change in the function $f(x)$ over this interval: $\int_1^5 f'(x)\,dx = f(5) - f(1)$. The graph shows that the function $f'(x)$ is a constant function, represented by the horizontal line $y = 4$. Therefore, $f'(x) = 4$. The area of the shaded region is given as 16 square units. Using the definite integral: $\int_1^5 4\,dx = [4x]_1^5 = 4(5) - 4(1) = 20 - 4 = 16$. This con.

33 [Intermediate] • Target: Choosing u and dv wisely (LIATE)

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

a. $\frac{1}{x-1} + \frac{1}{(x-1)^2}$ ✓

b. $\frac{x}{(x-1)^2}$

c. $\frac{1}{x-1}$

d. $\frac{2}{(x-1)^2}$

Applying Techniques of Integration principles, $x = (x-1) + 1 \Rightarrow A = 1, B = 1$. Therefore, the verified correct option is $\frac{1}{x-1} + \frac{1}{(x-1)^2}$.

34 [Intermediate] • Target: The recurring-integral trick

Arc length in parametric form ($x = x(t), y = y(t)$):

a. $\int_a^b \sqrt{(dx/dt)^2 + (dy/dt)^2}\,dt$ ✓

b. $\int_a^b \sqrt{(dx/dt) + (dy/dt)}\,dt$

c. $\int_a^b |dy/dx|\,dt$

d. $\sqrt{x(b)^2 + y(b)^2}$

Applying Applications of Integration principles, Parametric arc length. Therefore, the verified correct option is $\int_a^b \sqrt{(dx/dt)^2 + (dy/dt)^2}\,dt$.

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35 [Intermediate] • Target: Definite integrals by parts

Kinetic energy of rotating rod (length L, mass M, angular velocity ω):

a. $\frac{1}{3}M\omega^2L^2$

b. $\frac{1}{6}M\omega^2L^2$ ✓

c. $\frac{1}{2}M\omega^2L^2$

d. $M\omega^2L^2$

For a uniform rod rotated about one end, $I=\frac{1}{3}ML^2$, so the rotational kinetic energy is $K=\frac{1}{2}I\omega^2=\frac{1}{6}M\omega^2L^2$.

36 [Intermediate] • Target: Reduction formulas

Force on a flat plate submerged at depth d, width $w(y)$, density ρ:

a. $\rho g\int w(y)(d-y)\,dy$ ✓

b. $\rho g dw$

c. $\rho w d^2$

d. $\int w\,dy$

Applying Applications of Integration principles, Hydrostatic pressure × area. Therefore, the verified correct option is $\rho g\int w(y)(d-y)\,dy$.

37 [Intermediate] • Target: Integration by parts with inverse trig functions

Moments: $M_y=\rho\int_a^b x[f(x)-g(x)]dx$. Centroid $\bar{x}=M_y/M$ where $M=$

a. $\rho\int_a^b[f(x)-g(x)]dx$

b. $\rho(b-a)$

c. $\rho\cdot A$ **(same as first)** ✓

d. M_y

Applying Applications of Integration principles, $M=\rho A$ is equivalent. Therefore, the verified correct option is $\rho\cdot A$ (same as first).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A capstone mixed-technique problem • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Probability of event $a\le X\le b$ for PDF f:

a. $\int_a^b f(x)\,dx$ ✓

b. $f(b)-f(a)$

c. $f(a)\cdot(b-a)$

d. $F(b)\cdot F(a)$

Applying Applications of Integration principles, Probability = integral of PDF. Therefore, the verified correct option is $\int_a^b f(x)\,dx$.

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39 [Challenging] • Target: Visualizing an area found by integration by parts • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Centroid of semicircular region radius R, $y \geq 0$:

a. $\bar{y} = R/2$

b. $\bar{y} = 4R/(3\pi)$ ✓

c. $\bar{y} = 2R/3$

d. $\bar{y} = R/\pi$

Applying Applications of Integration principles, $\bar{y} = 4R/(3\pi)$. Therefore, the verified correct option is $\bar{y} = 4R/(3\pi)$.

40 [Challenging] • Target: Choosing between substitution and parts • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

A cylindrical tank of radius 3 m and height 2 m is full of water. Write the work to pump all the water to a point 1 m above the top as $W = k\rho g$. Find k.

a. 36π ✓

b. 54π

c. 36

d. 18π

A slab at height y has volume $9\pi\,dy$ and is lifted $3 - y$ m. $k = \int_0^2 9\pi(3-y)\,dy = 36\pi$.

41 [Challenging] • Target: The integration by parts formula • Cross-Lesson: 6.01 Related Rates

Work done lifting a 10 kg mass 4 m ($g = 10$ m/s²):

a. 400 **J** ✓

b. 40 J

c. 4 J

d. 100 J

Applying Applications of Integration principles, $W = mgh = 10 \cdot 10 \cdot 4 = 400$ J. Therefore, the verified correct option is 400 J.

42 [Challenging] • Target: Repeated and definite applications • Cross-Lesson: 6.02 Optimization and Curve Sketching

Volume of revolution of $y = x^3$ about x-axis from 0 to 1:

a. π

b. $\pi/5$

c. $\pi/7$ ✓

d. $3\pi/7$

Applying Applications of Integration principles, $V = \pi\int_0^1 x^6\,dx = \pi/7$. Therefore, the verified correct option is $\pi/7$.

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43 [Challenging] • Target: Choosing u and dv wisely (LIATE) • Cross-Lesson: 7.01 Integration by Substitution

Moment of inertia of disk radius R, mass M about its axis:

a. MR^2

b. $\frac{1}{4}MR^2$

c. $\frac{1}{2}MR^2$ ✓

d. $2MR^2$

Applying Applications of Integration principles, $I = \frac{1}{2}MR^2$. Therefore, the verified correct option is $\frac{1}{2}MR^2$.

44 [Challenging] • Target: The recurring-integral trick • Cross-Lesson: 8.01 Area Between Curves

Find the angle between $\mathbf{u} = \langle 1, 2, -4 \rangle$ and $\mathbf{v} = \langle 4, -2, 0 \rangle$.

a. $\frac{\pi}{4}$

b. $\frac{\pi}{3}$

c. $\frac{\pi}{6}$

d. $\frac{\pi}{2}$ ✓

$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = \frac{0}{2\sqrt{105}} = 0$, so $\theta = \frac{\pi}{2}$.

45 [Challenging] • Target: Definite integrals by parts • Cross-Lesson: 8.02 Volumes of Revolution

Spring: force $F = kx$. Work stretching from 0 to d:

a. $kd^2/2$ ✓

b. kd

c. kd^2

d. $2kd$

Applying Applications of Integration principles, $W = \int_0^d kx\,dx = kd^2/2$. Therefore, the verified correct option is $kd^2/2$.

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